

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$([g,h])^\circ = (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$$

the converse of a fold over decimals undoes the algebra $[g,h]$ and then either **wrap**, or the fold's converse on the front and **snoc**

1	$([g,h])^\circ$	$(\emptyset) = [\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}(\emptyset)) \emptyset$
2	$= ([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}([g,h])) [g,h])^\circ$	$(R + S) [P,Q] = [R P, S Q]$
3	$= ([\text{wrap}, \text{snoc}]^\circ [g, \neg \text{Digit}([g,h]) h])^\circ$	$[U,V]^\circ [R,S] = (U^\circ R) \cup (V^\circ S)$
4	$= ((\text{wrap}^\circ g) \cup (\text{snoc}^\circ \neg \text{Digit}([g,h]) h))^\circ$	$(R \cup S)^\circ = S^\circ \cup R^\circ$
5	$= (\text{snoc}^\circ \neg \text{Digit}([g,h]) h)^\circ \cup (\text{wrap}^\circ g)^\circ$	$R \cup S = S \cup R$
6	$= (\text{wrap}^\circ g)^\circ \cup (\text{snoc}^\circ \neg \text{Digit}([g,h]) h)^\circ$	$(R S)^\circ = S^\circ R^\circ$
7	$= (g^\circ \text{ wrap}) \cup ((\neg \text{Digit}([g,h]) h)^\circ \text{ snoc})$	$(R S)^\circ = S^\circ R^\circ$
8	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ \text{ snoc})$	$\neg \text{Digit}(R)^\circ = \neg \text{Digit}(R^\circ)$
9	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$	

$\text{val}^\circ = (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{ snoc})$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

1	$\text{op}^\circ(m,p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$ op° splits a number into its quotient and remainder by 10
2	$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m,p)) \iff 10 \leq m$ op° gives a pair with a nonzero first component exactly at the numbers with two or more digits
3	$(\exists d. \text{embed}^\circ(m,d)) \iff 0 < m \wedge m < 10$ embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

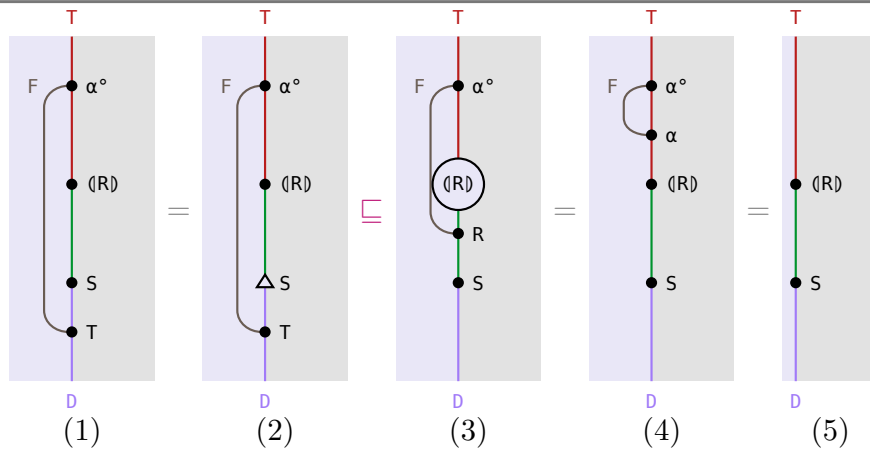
(6.2b)

the law	what it says
1 $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
2 $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
3 $\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
4 $X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

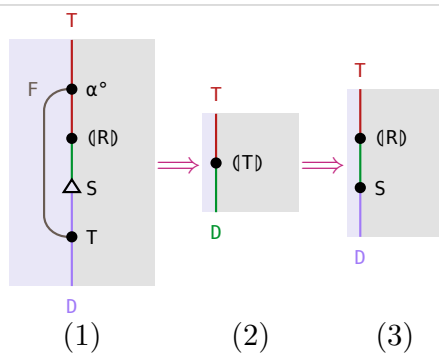
$$F(S) \ T \sqsubseteq R \ S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle \ S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



1 $\alpha^\circ F(\langle R \rangle \ S) \ T$	
2 $= \alpha^\circ F(\langle R \rangle) \ F(S) \ T$	$F(S) \ T \sqsubseteq R \ S$
3 $\sqsubseteq \alpha^\circ F(\langle R \rangle) \ R \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
4 $= \alpha^\circ \alpha \ \langle R \rangle \ S$	$\alpha^\circ \alpha = 1$
5 $= \langle R \rangle \ S$	



1 $F(S) \ T \sqsubseteq R \ S$	$F(S) \ T \sqsubseteq R \ S$ $\Rightarrow \alpha^\circ F(\langle R \rangle \ S) \ T \sqsubseteq \langle R \rangle \ S$
2 $\Rightarrow \alpha^\circ F(\langle R \rangle \ S) \ T \sqsubseteq \langle R \rangle \ S$	$\alpha^\circ F(X) \ R \sqsubseteq X$ $\Rightarrow \langle R \rangle \sqsubseteq X$
3 $\Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle \ S$	

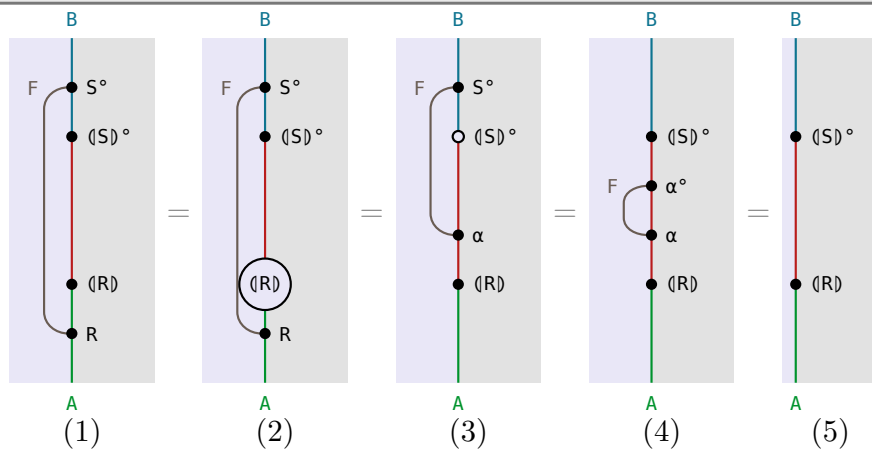
$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$ (6.2d)	
if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T	
1	$\langle R \rangle \ S$
2	$= \langle R \rangle \ S$
3	$= \alpha^\circ \ \alpha \ \langle R \rangle \ S$
4	$= \alpha^\circ \ F(\langle R \rangle) \ R \ S$
5	$\sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$
	$\alpha^\circ \ \alpha = 1$ $\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$ $R \ S \sqsubseteq F(S) \ T$
1	$R \ S \sqsubseteq F(S) \ T$
2	$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$
3	$\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$
	$R \ S \sqsubseteq F(S) \ T$ $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$ $X \sqsubseteq \alpha^\circ \ F(X) \ R$ $\Rightarrow X \sqsubseteq \langle R \rangle$

§6.3. Hylomorphisms

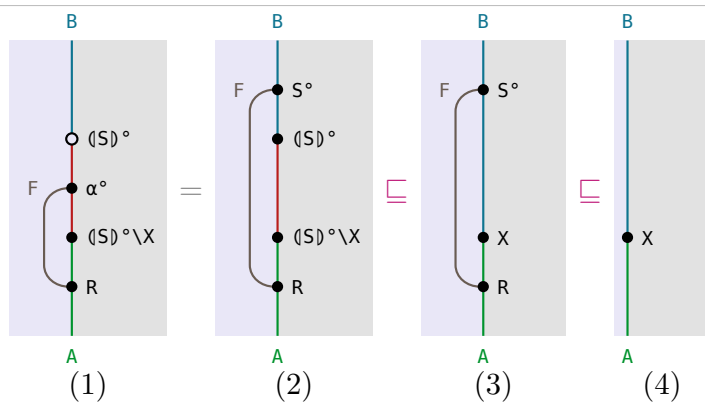
$$(\mathbb{S})^\circ (\mathbb{R}) = (\mu X : S^\circ F(X) R)$$

(6.3a)

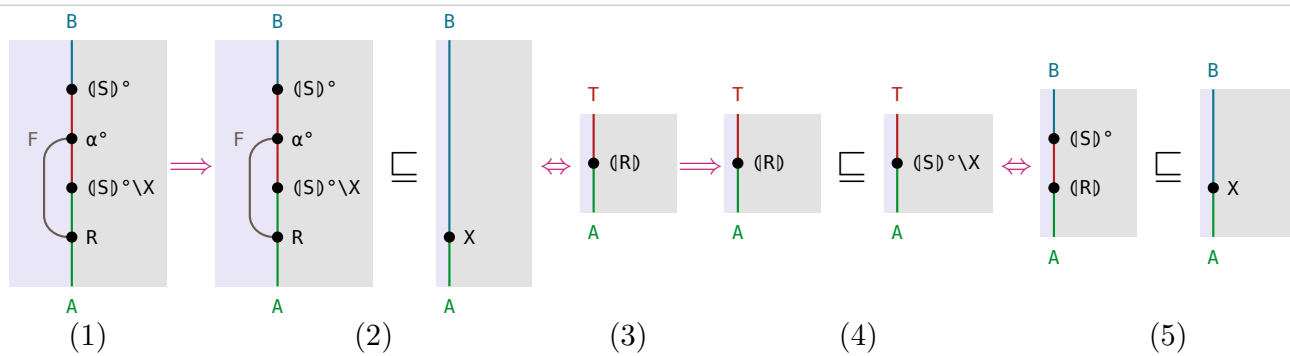
hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



1	$S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R$	
2	$= S^\circ F((\mathbb{S})^\circ) F((\mathbb{R})) R$	$\alpha (\mathbb{R}) = F((\mathbb{R})) R$
3	$= S^\circ F((\mathbb{S})^\circ) \alpha (\mathbb{R})$	$(\mathbb{R})^\circ \alpha^\circ = R^\circ F((\mathbb{R})^\circ)$
4	$= (\mathbb{S})^\circ \alpha^\circ \alpha (\mathbb{R})$	$\alpha^\circ \alpha = 1$
5	$= (\mathbb{S})^\circ (\mathbb{R})$	



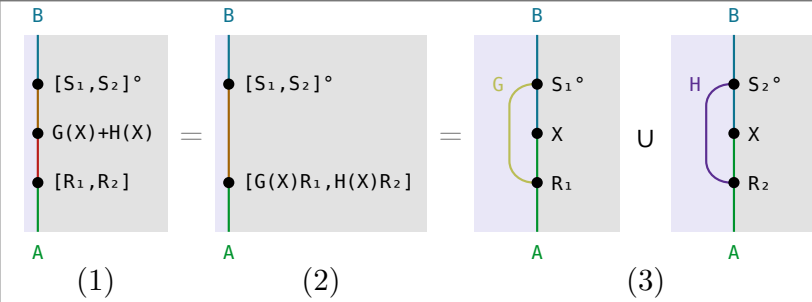
1	$(\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R$	$(\mathbb{R})^\circ \alpha^\circ = R^\circ F((\mathbb{R})^\circ)$
2	$= S^\circ F((\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X)) R$	$S (S \setminus R) \sqsubseteq R$
3	$\sqsubseteq S^\circ F(X) R$	$S^\circ F(X) R \sqsubseteq X$
4	$\sqsubseteq X$	



1	$S^\circ F(X) R \sqsubseteq X$	$S^\circ F(X) R \sqsubseteq X$
2	$\Rightarrow (\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \sqsubseteq X$	$\Rightarrow (\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \sqsubseteq X$
3	$\Leftrightarrow \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \sqsubseteq (\mathbb{S})^\circ \setminus X$	$X \sqsubseteq S \setminus R \Leftrightarrow S X \sqsubseteq R$
4	$\Rightarrow (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ \setminus X$	$\alpha^\circ F(X) R \sqsubseteq X$
5	$\Leftrightarrow (\mathbb{S})^\circ (\mathbb{R}) \sqsubseteq X$	$\Rightarrow (\mathbb{R}) \sqsubseteq X$

$$([S_1, S_2])^\circ ([R_1, R_2]) = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

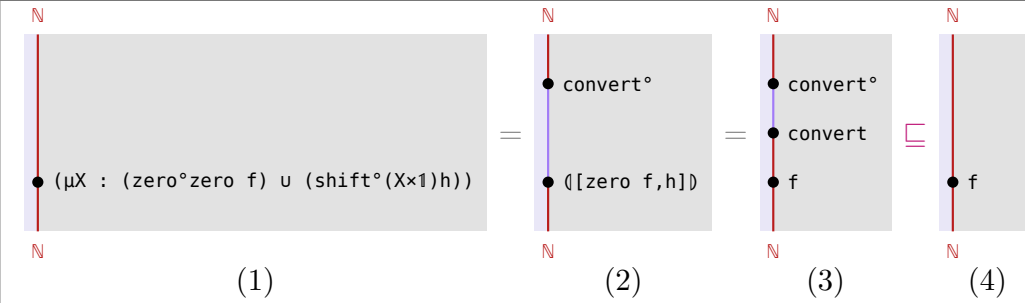


1	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$(R + S) [P, Q] = [R P, S Q]$
2	$= [S_1, S_2]^\circ [G(X) R_1, H(X) R_2]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
3	$= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$	

§6.4. Fast exponentiation and modulus computation

$$\text{shift } f = (f \times 1) h \Rightarrow (\mu X : (\text{zero}^\circ \text{zero } f) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f \quad (6.4a)$$

once f turns **shift** into h , the recursion that halves the argument at each step computes f



1	$(\mu X : (\text{zero}^\circ \text{zero } f) \cup (\text{shift}^\circ (X \times 1) h))$	$\text{convert}^\circ ([g, h]) = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$
2	$= \text{convert}^\circ ([\text{zero } f, h])$	$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h$
3	$= \text{convert}^\circ \text{convert } f$	$\Rightarrow \text{convert } f = ([g, h])$
4	$\sqsubseteq f$	$\text{Simple}(\text{convert})$

$$(\mu X : (\text{zero}^\circ \text{one}) \cup (\text{shift}^\circ (X \times 1) \text{op}(a))) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes a to the power b

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d

$$(\mu X : (\text{zero}^\circ \text{zero}) \cup (\text{shift}^\circ (X \times 1) \text{op}(b))) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves a at each step computes the remainder of a by b

$\text{zero } \text{mod}(b) = \text{zero}$

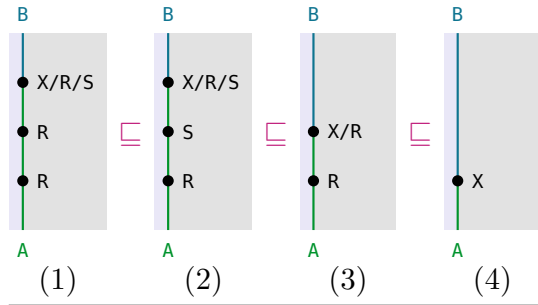
$0 \text{ mod } b$ is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d

§6.5. Unique fixed points

$$\begin{aligned}
 & \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{X} & (6.5a) \\
 & ((\mathbf{X}/\mathbf{R})/\mathbf{S}) \text{ followed by } \mathbf{R}\mathbf{R} \text{ stays below } \mathbf{X} \text{ when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \\
 & \mathbf{inductive}(\mathbf{S}) \wedge \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow \mathbf{inductive}(\mathbf{R}) \\
 & \mathbf{R} \text{ is inductive when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \text{ for an inductive } \mathbf{S}
 \end{aligned}$$



1	$(\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R}$	$\mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R}$
2	$\sqsubseteq (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{S} \mathbf{R}$	$(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$
3	$\sqsubseteq (\mathbf{X} / \mathbf{R}) \mathbf{R}$	$(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$
4	$\sqsubseteq \mathbf{X}$	

$$\mathbf{member} = \emptyset \quad (6.5b)$$

a constant relator records no elements, so its membership is empty

$$\mathbf{member} = [\mathbf{member}, \mathbf{member}]$$

a member of $\mathbf{F}+\mathbf{G}$ is a member of whichever summand the value is in

$$\mathbf{member} = (\pi_1 \mathbf{member}) \cup (\pi_2 \mathbf{member})$$

a member of $\mathbf{F} \times \mathbf{G}$ is a member of either component

$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{X} = \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \wedge \mathbf{Y} = \mathbf{S} \mathbf{F}(\mathbf{Y}) \mathbf{R} \Rightarrow \mathbf{X} = \mathbf{Y} & (6.5c) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive, two solutions } \mathbf{X}, \mathbf{Y} \text{ of } \mathbf{X} = \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ are equal}
 \end{aligned}$$

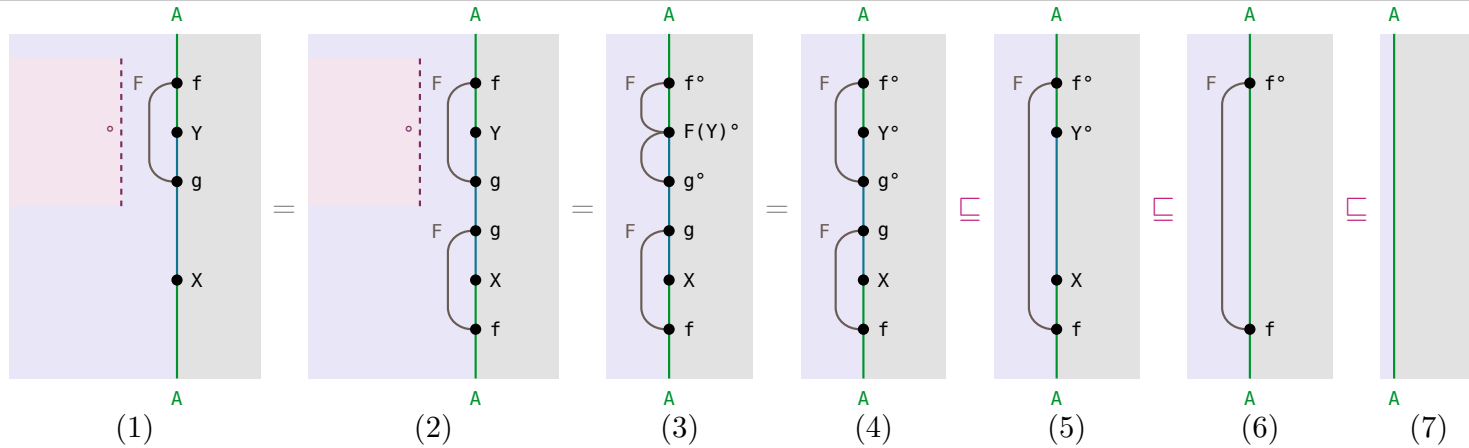
$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{Entire}(\mathbf{S}) \wedge \mathbf{Entire}(\mathbf{R}) \wedge \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \sqsubseteq \mathbf{X} \Rightarrow \mathbf{Entire}(\mathbf{X}) & (6.5d) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive and } \mathbf{S}, \mathbf{R} \text{ are entire, a pre-fixed point of } \mathbf{X} \mapsto \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ is entire}
 \end{aligned}$$

$$\mathbf{inductive}(\mathbf{g} \mathbf{member}) \wedge \mathbf{Map}(\mathbf{g}) \wedge \mathbf{Map}(\mathbf{f}) \wedge \mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \Rightarrow \mathbf{Map}(\mathbf{X}) \quad (6.5e)$$

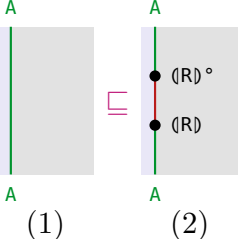
when $\mathbf{g} \mathbf{member}$ is inductive and $\mathbf{f}, \mathbf{g}$ are maps, the solution of $\mathbf{X} = \mathbf{g}\mathbf{F}(\mathbf{X})\mathbf{f}$ is a map

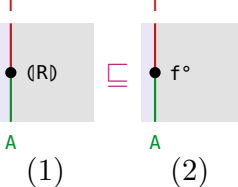
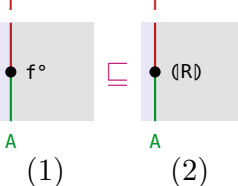
$$\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \wedge \mathbf{Simple}(\mathbf{g}) \wedge \mathbf{Simple}(\mathbf{f}) \wedge \mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1} \Rightarrow (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X} \sqsubseteq \mathbf{1}$$

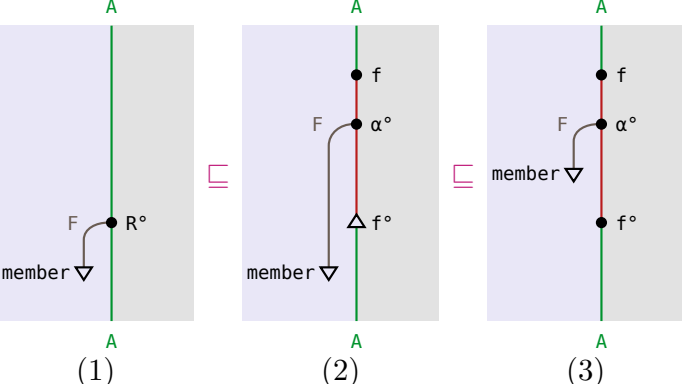
$(\mathbf{g}\mathbf{F}(\mathbf{Y})\mathbf{f})^\circ$ followed by $\mathbf{X}$ stays below $\mathbf{1}$ when $\mathbf{g}, \mathbf{f}$ are simple and $\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$



1	$(\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X}$	$\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$
2	$= (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$(\mathbf{R} \mathbf{S})^\circ = \mathbf{S}^\circ \mathbf{R}^\circ$
3	$= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y})^\circ \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$\mathbf{PreservesRecip}(\mathbf{F})$
4	$= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ) \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$\mathbf{Simple}(\mathbf{g})$
5	$\sqsubseteq \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ \mathbf{X}) \mathbf{f}$	$\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$
6	$\sqsubseteq \mathbf{f}^\circ \mathbf{f}$	$\mathbf{Simple}(\mathbf{f})$
7	$\sqsubseteq \mathbf{1}$	

$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \implies 1 \sqsubseteq (R)^\circ (R)$ (6.5f)	
when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective	
 (1) (2)	
1	1
2	$\sqsubseteq (R)^\circ (R)$
$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S \subseteq F(X) \subseteq X \implies \text{Entire}(X) \text{ at } S=R^\circ$	

$\text{Map}(f) \wedge R f \subseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \implies f^\circ = (R)$ (6.5g)	
a map f with $Rf \subseteq F(f)\alpha$, for a surjective R , has the fold (R) as its converse	
 (1) (2)	
1	(R)
2	$\sqsubseteq f^\circ$
shunting f , fusion, $Rf \subseteq F(f)\alpha$	
 (1) (2)	
1	f°
2	$\sqsubseteq (R)$
$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \implies 1 \sqsubseteq (R)^\circ (R) \text{ and } (R)f \sqsubseteq 1$	

$\text{Map}(f) \wedge R f \subseteq F(f) \alpha \implies R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ$ (6.5h)	
$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is	
 (1) (2) (3)	
1	$R^\circ \text{ member}$
2	$\sqsubseteq f \alpha^\circ F(f^\circ) \text{ member}$
3	$\sqsubseteq f \alpha^\circ \text{ member } f^\circ$
$\text{Map}(f) \wedge R f \subseteq F(f) \alpha \implies R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$	
$F(R) \text{ member} \sqsubseteq \text{member } R$	

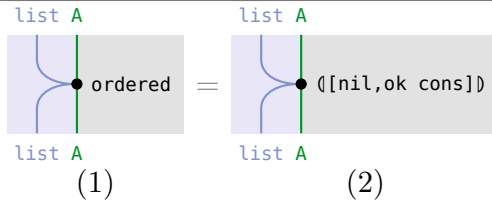
§6.6. Sorting by selection

(6.6a)			
	name	type	definition
1	orderedP(R)	P[A]	$\text{orderedP}[] = \text{true}, \text{orderedP}([a] \# xs) = (\forall x. x \in xs \implies R(a, x)) \wedge \text{orderedP}(xs)$
2	ordered	[A] → [A]	$xs \text{ ordered } ys \iff xs = ys \wedge \text{orderedP}(xs)$
3	ok	A × [A] → A × [A]	$(a, xs) \text{ ok } q \iff (a, xs) = q \wedge (\forall x. x \in xs \implies R(a, x))$
			meaning
			every element is R-below each element after it
			the coreflexive passing exactly the lists orderedP holds of
			the coreflexive passing (a, x) when a is R-below every element of x

$$\text{ordered} = ([\text{nil}, \text{ok cons}])$$

(6.6b)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

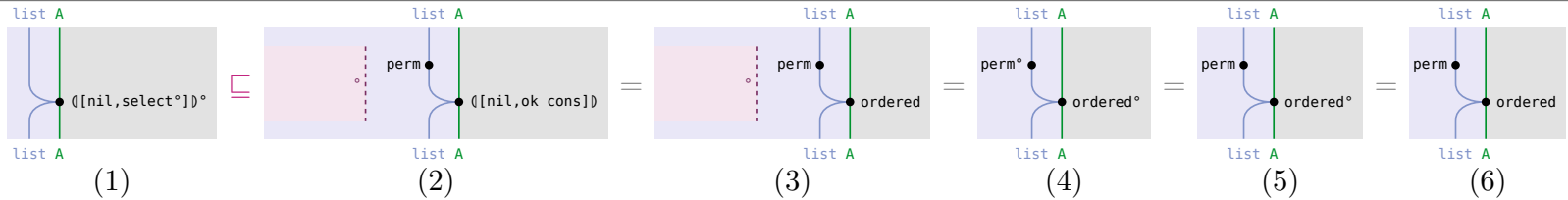


1	ordered	fold uniqueness
2	$= ([\text{nil}, \text{ok cons}])$	

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered}$$

(6.6c)

every output of unfolding the input by **select** is a sorted permutation of it

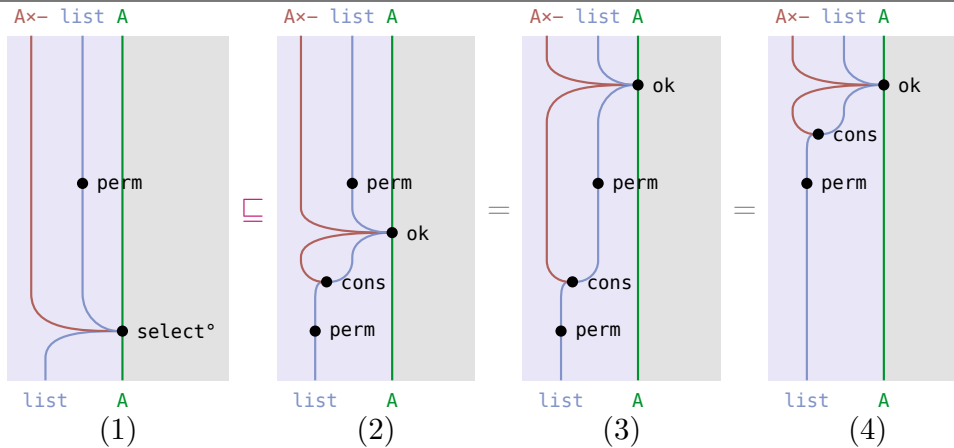


1	$([\text{nil}, \text{select}^\circ])^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
2	$\sqsubseteq ([\text{nil}, \text{ok cons}]) \text{ perm}^\circ$	$\Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq ([\text{nil}, \text{ok cons}]) \text{ perm}^\circ$
3	$= (\text{ordered perm})^\circ$	(6.6b) $\text{ordered} = ([\text{nil}, \text{ok cons}])$
4	$= \text{perm}^\circ \text{ ordered}^\circ$	$(R S)^\circ = S^\circ R^\circ$
5	$= \text{perm ordered}^\circ$	$\text{perm}^\circ = \text{perm}$
6	$= \text{perm ordered}$	$\text{ordered}^\circ = \text{ordered}$

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{ select}^\circ \sqsubseteq \text{ok cons perm}$$

(6.6d)

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation

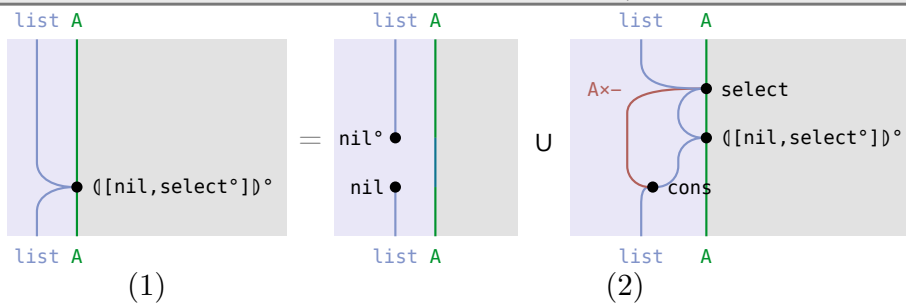


1	$(1 \times \text{perm}) \text{ select}^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
2	$\sqsubseteq (1 \times \text{perm}) \text{ ok cons perm}$	$\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ ok}$
3	$= \text{ok } (1 \times \text{perm}) \text{ cons perm}$	$\text{cons perm} = (1 \times \text{perm}) \text{ cons perm}$
4	$= \text{ok cons perm}$	

$$([\text{nil}, \text{select}^\circ])^\circ = (\text{nil}^\circ \text{ nil}) \cup (\text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons})$$

(6.6e)

selection sort returns **[]** on **[]**, and otherwise selects **(a,y)**, sorts **y** and conses **a** back on

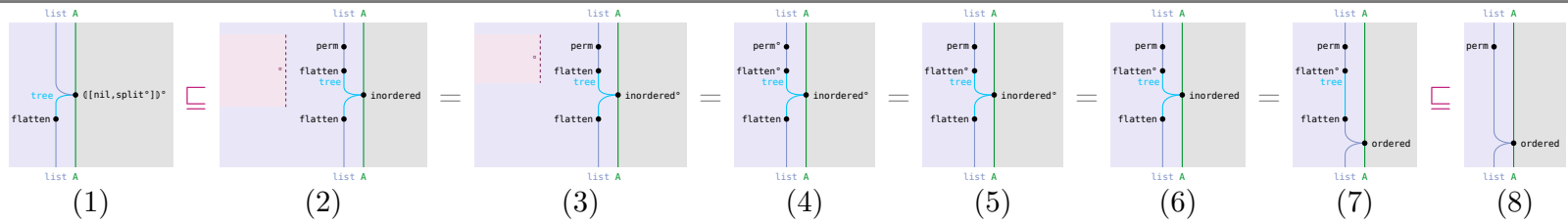


1	$([\text{nil}, \text{select}^\circ])^\circ$	unfolding the converse of a fold
2	$= (\text{nil}^\circ \text{ nil}) \cup (\text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons})$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \quad (6.6f)$$

flatten $\sqsubseteq$ perm ordered

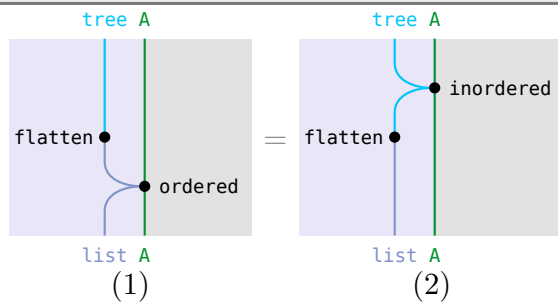
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



1	$[\text{nil}, \text{split}^\circ]^\circ \text{ flatten}$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$
2	$\sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$	$\Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} \sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$
3	$= (\text{flatten perm})^\circ \text{ inordered}^\circ \text{ flatten}$	$(R S)^\circ = S^\circ R^\circ$
4	$= \text{perm}^\circ \text{ flatten}^\circ \text{ inordered}^\circ \text{ flatten}$	$(R S)^\circ = S^\circ R^\circ$
5	$= \text{perm flatten}^\circ \text{ inordered}^\circ \text{ flatten}$	$\text{perm}^\circ = \text{perm}$
6	$= \text{perm flatten}^\circ \text{ inordered flatten}$	$\text{inordered}^\circ = \text{inordered}$
7	$= \text{perm flatten}^\circ \text{ flatten ordered}$	$(6.6g) (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c))$
8	$\sqsubseteq \text{perm ordered}$	$\Rightarrow \text{flatten ordered} = \text{inordered flatten}$
		$\text{flatten}^\circ \text{ flatten} \sqsubseteq \mathbb{1}$

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6g)$$

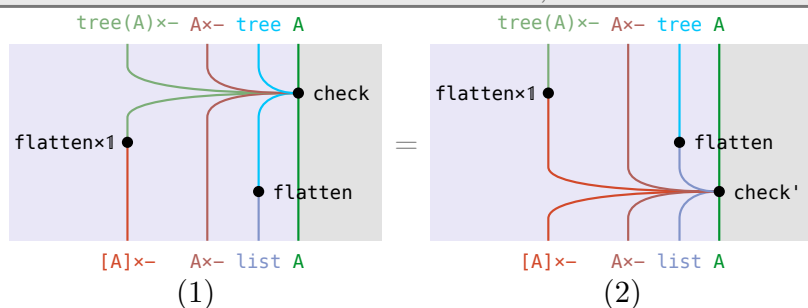
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



1	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
2	$= \text{inordered flatten}$	

$$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \quad (6.6h)$$

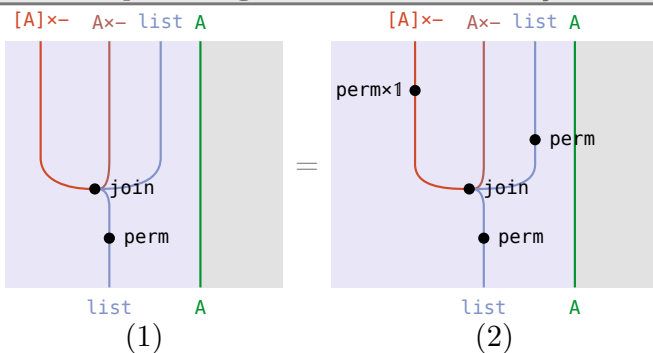
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



1	$\text{check} (\text{flatten} \times (1 \times \text{flatten}))$	$b \text{ intree } x \Leftrightarrow b \text{ inlist } \text{flatten}(x)$
2	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$	

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm} \quad (6.6i)$$

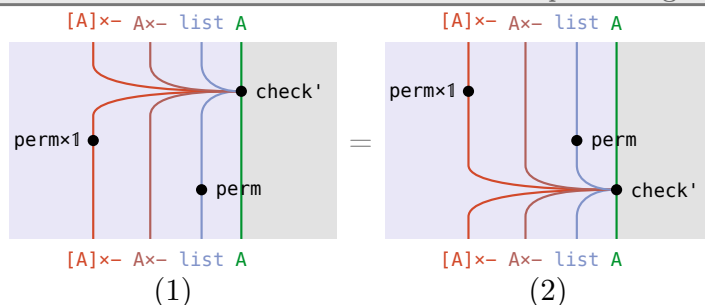
permuting the list $x \mathrel{++} [a] \mathrel{++} y$ is the same as first permuting x and y and then permuting the joined list



1	join perm	perm is transitive and closed under $++$
2	$= (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$	

$$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}' \quad (6.6j)$$

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

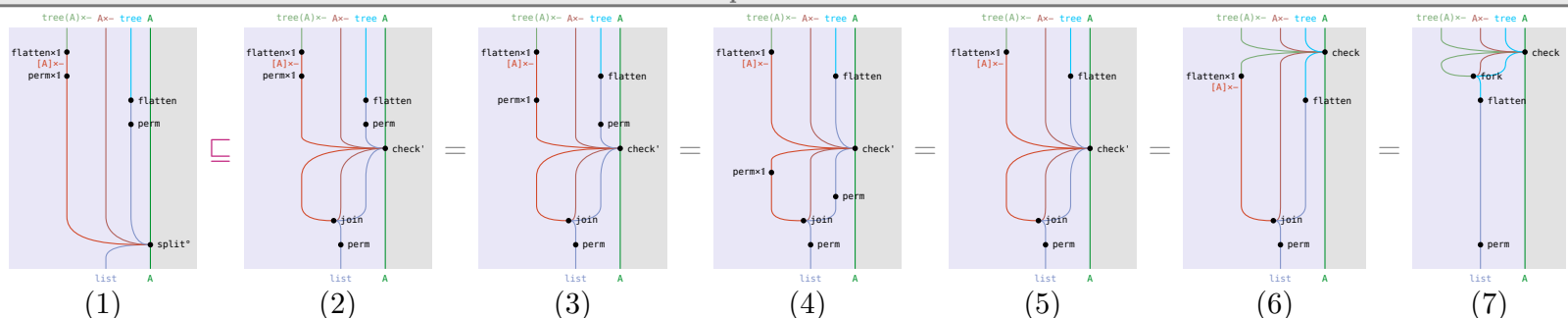


1	$\text{check}' (\text{perm} \times (1 \times \text{perm}))$	$b \text{ inlist } x \iff b \text{ inlist } \text{perm}(x)$
2	$= (\text{perm} \times (1 \times \text{perm})) \text{ check}'$	

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ \sqsubseteq \text{check}' \text{ fork} \quad (6.6k)$$

flatten perm

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted

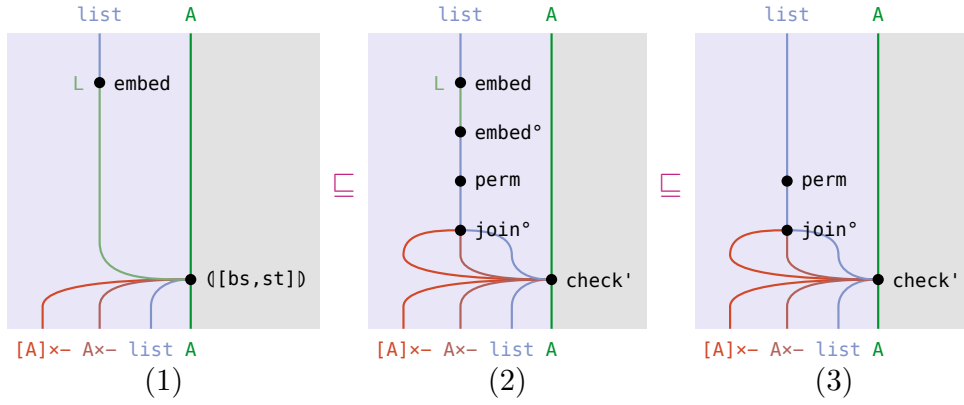


1	$((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$
2	$\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ check}' \text{ join perm}$	
3	$= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{ check}' \text{ join perm}$	(6.6j) $\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}'$
4	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$	(6.6i) $\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$
5	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \text{ join perm}$	(6.6h) $\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$
6	$= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{ join perm}$	$\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{ join}$
7	$= \text{check fork flatten perm}$	

bs $\sqsubseteq$ wrap perm join $^\circ$ check' $\wedge$ ($1 \times$ (perm join $^\circ$ check')) st $\sqsubseteq$ cons perm join $^\circ$ check' $\Rightarrow$ embed (6.6l)

$([bs, st]) \sqsubseteq$ perm join $^\circ$ check'

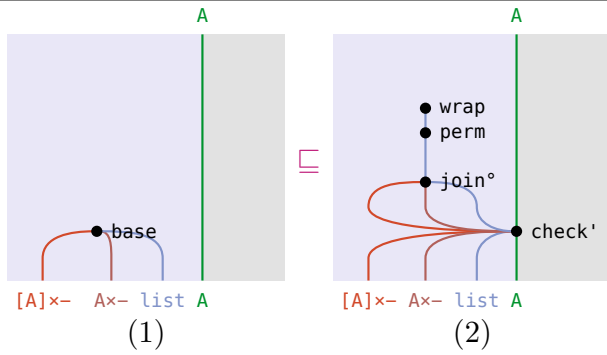
turning a list into a non-empty list and folding it with any $[bs, st]$ meeting the two conditions below gives only **checked triples** (x, a, y) whose join $x \# [a] \# y$ is a permutation of the list



1	embed $([bs, st])$	$bs \sqsubseteq$ wrap perm join $^\circ$ check' $\wedge$ ($1 \times$ (perm join $^\circ$ check')) st $\sqsubseteq$ cons perm join $^\circ$ check' $\Rightarrow$ embed $([bs, st]) \sqsubseteq$ embed embed $^\circ$ perm join $^\circ$ check'
2	$\sqsubseteq$ embed embed $^\circ$ perm join $^\circ$ check'	embed embed $^\circ$ perm join $^\circ$ check' $\sqsubseteq$ perm join $^\circ$ check'
3	$\sqsubseteq$ perm join $^\circ$ check'	

base $\sqsubseteq$ wrap perm join $^\circ$ check' (6.6m)

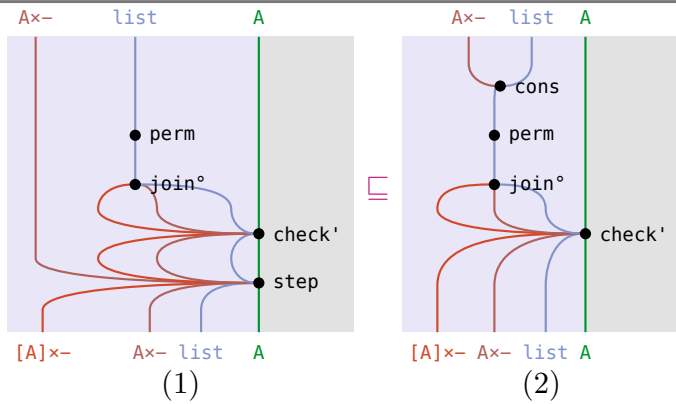
base(a) is a **checked triple** whose join is a permutation of the one-element list $[a]$



1	base	$[] \# [a] \# [] = [a]$, and check' holds on empty lists
2	$\sqsubseteq$ wrap perm join $^\circ$ check'	

connected(R) $\Rightarrow$ ($1 \times$ (perm join $^\circ$ check')) step $\sqsubseteq$ cons perm join $^\circ$ check' (6.6n)

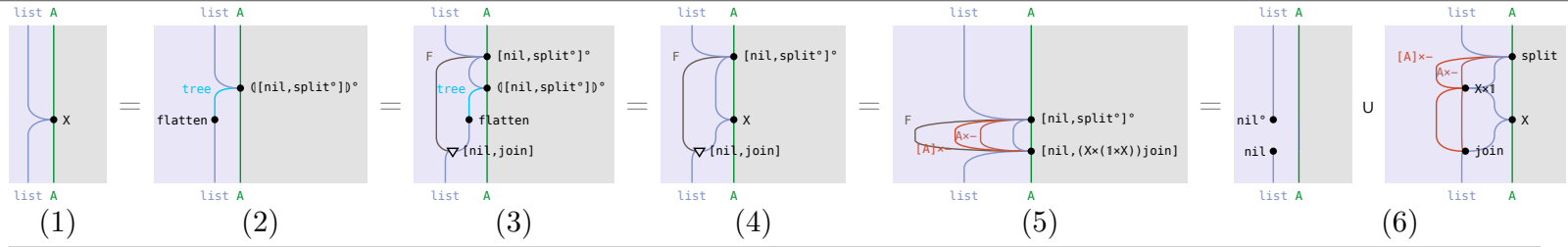
if (x, b, y) is a **checked triple** for a permutation of $\mathbf{l}$, then **step(a, (x, b, y))** is a **checked triple** for a permutation of $[a] \# \mathbf{l}$



1	$(1 \times$ (perm join $^\circ$ check')) step	aRb puts a in front of x; otherwise bRa and a goes in front of y
2	$\sqsubseteq$ cons perm join $^\circ$ check'	

$$X = ([nil, split^\circ])^\circ \text{ flatten} \Rightarrow X = (nil^\circ nil) \cup (split (X \times (1 \times X)) \text{ join}) \quad (6.6o)$$

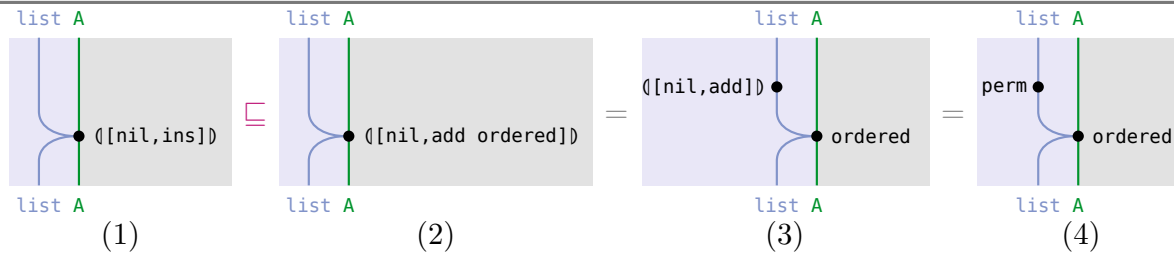
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



1	X	$X = ([nil, split^\circ])^\circ \text{ flatten}$
2	$= ([nil, split^\circ])^\circ \text{ flatten}$	$S^\circ F([S]^\circ [R]) R = [S]^\circ [R]$
3	$= [nil, split^\circ]^\circ F([nil, split^\circ])^\circ \text{ flatten} [nil, join]$	$X = ([nil, split^\circ])^\circ \text{ flatten}$
4	$= [nil, split^\circ]^\circ F(X) [nil, join]$	$F(S) [T, U] = [T, (S \times (1 \times S)) U]$
5	$= [nil, split^\circ]^\circ [nil, (X \times (1 \times X)) \text{ join}]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
6	$= (nil^\circ nil) \cup (split (X \times (1 \times X)) \text{ join})$	

$$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow ([nil, ins])^\circ \sqsubseteq \text{perm ordered} \quad (6.6p)$$

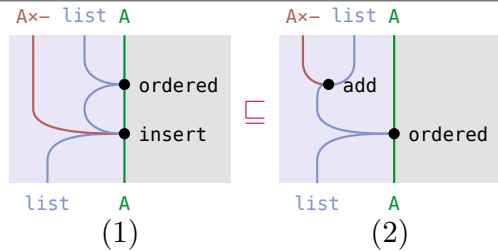
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



1	$([nil, ins])^\circ$	$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$
2	$\sqsubseteq ([nil, add \text{ ordered}])^\circ$	$\Rightarrow ([nil, ins])^\circ \sqsubseteq ([nil, add \text{ ordered}])^\circ$
3	$= ([nil, add])^\circ \text{ ordered}$	$[nil, add]^\circ \text{ ordered} = ([nil, add \text{ ordered}])^\circ$
4	$= \text{perm ordered}$	$\text{perm} = ([nil, add])^\circ$

$$\text{connected}(R) \wedge (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{ insert} \sqsubseteq \text{add ordered} \quad (6.6q)$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



1	$(1 \times \text{ordered}) \text{ insert}$	insert splices a in, and R transitive and connected
2	$\sqsubseteq \text{add ordered}$	

§6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***

$$1 \sqsubseteq X \wedge X X \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

X contains **R** exactly when it contains **R***

$$1 \sqsubseteq R^* \quad (6.7c)$$

R* is reflexive

1	1	$R \sqsubseteq R \cup S$
2	$\sqsubseteq 1 \cup (R R^*)$	$1 \cup (R R^*) = R^*$
3	$= R^*$	

$R \subseteq R^*$ (6.7d)	
R^* contains R	
1	R
2	$= R$ (6.7c) $1 \subseteq R^*$
3	$\subseteq R R^*$ $R R^* \subseteq R^*$
4	$\subseteq R^*$

$(1 \cup (R (R^* / R^*))) R^* \subseteq R^*$ (6.7e)	
R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$, stated through division	
$R^* R^* \subseteq R^*$	
R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$	
1	$(1 \cup (R (R^* / R^*))) R^*$ $(S \cup T) R = (S R) \cup (T R)$
2	$= (R^*) \cup (R (R^* / R^*) R^*)$
3	$= R^* \cup (R (R^* / R^*) R^*)$ $(R / R) R \subseteq R$
4	$\subseteq R^* \cup (R R^*)$ $R R^* \subseteq R^*$
5	$\subseteq R^*$

$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow 1 \cup (R X) \subseteq X$ (6.7f)	
a reflexive transitive X containing R is a prefixed point of $Y \mapsto 1 \cup RY$	
$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow R^* \subseteq X$	
a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*	
1	$1 \cup (R X)$ $1 \subseteq X$
2	$\subseteq X \cup (R X)$ $R \subseteq X$
3	$\subseteq X \cup (X X)$
4	$\subseteq X$

$\frac{R^*}{\exists} = (\frac{1}{\exists}, \frac{R R^*}{\exists}) \text{ cup}$ (6.7g)	
the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors	
1	$\frac{R^*}{\exists}$ $1 \cup (R R^*) = R^*$
2	$= \frac{1 \cup (R R^*)}{\exists}$ $\frac{R \cup S}{\exists} = (\frac{R}{\exists}, \frac{S}{\exists}) \text{ cup}$
3	$= (\frac{1}{\exists}, \frac{R R^*}{\exists}) \text{ cup}$

$R - \emptyset = R$ (6.7h)	
taking nothing away leaves R	
$R \cup (S - R) = R \cup S$	
beside R , only the part of S outside R adds anything	
$R - (S \cup T) = R - S - T$	
taking away S and then T takes away $S \cup T$	
$(R \cup S) - T = (R - T) \cup (S - T)$	
taking T away from a union takes it away from each part	

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \Rightarrow (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$ (6.7i)	
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ	

$S R^* = (\mu X : S \cup (X R))$ (6.7j)	
S followed by any number of R steps is the least X containing S and closed under a further R step	
$R^* S = (\mu X : S \cup (R X))$	
any number of R steps followed by S is the least X containing S and closed under an R step in front	

$\theta(\emptyset, S) = S R^*$ (6.7k)		
started with nothing found and S to explore, θ computes SR^*		
1	$\theta(\emptyset, S)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2	$\triangleq \emptyset \cup (\mu X : S \cup ((X R) - \emptyset))$	(6.7h) $R - \emptyset = R$
3	$= \emptyset \cup (\mu X : S \cup (X R))$	$\emptyset \cup R = R$
4	$= (\mu X : S \cup (X R))$	(6.7j) $S R^* = (\mu X : S \cup (X R))$
5	$= S R^*$	

$\theta(P, \emptyset) = P$ (6.7l)		
with nothing left to explore, θ returns what it has found		
1	$\theta(P, \emptyset)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2	$\triangleq P \cup (\mu X : \emptyset \cup ((X R) - P))$	$(\mu X : \emptyset \cup ((X R) - P)) = \emptyset$
3	$= P \cup \emptyset$	$R \cup \emptyset = R$
4	$= P$	